
Project

PeanutIQ

Agentic AI Powered Seed-to-Harvest

Smart Peanut Farming Ecosystem

& End-to-End Marketplace

1. Introduction

Agriculture is the backbone of Pakistan's economy, and peanut is one of the most important cash crops of the Pothwar region. Pakistan cultivates peanuts across approximately 336,767 acres of land, producing nearly 141,693 metric tonnes annually, with around 85 percent of production concentrated in Attock, Chakwal, Talagang, and Rawalpindi. Despite its economic importance, peanut farming remains largely dependent on traditional practices, fragmented advisory systems, and limited access to modern digital technologies.

Farmers face multiple challenges throughout the crop lifecycle, beginning with seed selection and continuing through crop management, disease control, and harvest planning. Most farmers lack reliable and affordable methods for assessing seed quality before planting, resulting in poor germination rates and reduced crop establishment. Similarly, peanut crops remain vulnerable to several diseases, including Early Leaf Spot, Late Leaf Spot, Collar Rot, and Crown Rot, which are often detected only after significant damage has occurred. In addition, farmers frequently rely on manual observation, personal experience, and delayed expert consultation, leading to inefficient decision-making and avoidable crop losses.

Although advancements in Artificial Intelligence, Computer Vision, Deep Learning, and Agentic AI have created new opportunities for intelligent agriculture, existing agricultural solutions typically focus on isolated functions such as disease detection, advisory services, or monitoring systems. Farmers, however, require a unified and continuous support system capable of assisting them throughout the complete crop lifecycle.

Another major challenge is the gap between agricultural research and practical farming.

Decades of valuable peanut research conducted by the National Agricultural Research Centre (NARC) Islamabad and the Barani Agricultural Research Institute (BARI) Chakwal, including information on seed varieties, disease management, and cultivation practices, often remains confined to research institutions and does not effectively reach farmers at the field level. Similarly, farmers have limited access to market awareness, buyer information, and data-driven decision support.

To address these challenges, this project proposes PeanutIQ, an Agentic AI Powered Seed-to-Harvest Smart Peanut Farming Ecosystem. The system is built around three core pillars: Seed Intelligence, Disease Intelligence, and an Agentic AI Farming Companion. The Seed Intelligence module utilizes computer vision techniques to assess seed quality, estimate germination potential, and support informed planting decisions. The Disease Intelligence module enables early disease detection, severity assessment, disease progression prediction, outbreak risk analysis, and

preventive guidance using deep learning and predictive analytics. . The Agentic AI Farming Companion acts as an intelligent agricultural assistant that combines farmer history, crop information, disease analysis, and trusted agricultural knowledge to generate personalized, explainable, and context-aware recommendations throughout the crop lifecycle.

The platform is being developed in collaboration with the National Agricultural Research Centre (NARC) Islamabad and the Barani Agricultural Research Institute (BARI) Chakwal, ensuring that research-backed agricultural knowledge is transformed into practical and accessible guidance for farmers. Future enhancements of the ecosystem may include market intelligence, buyer connectivity, agricultural marketplaces, and additional crop support, enabling PeanutIQ to evolve into a comprehensive digital agriculture platform for Pakistan.

2. Related System Analysis / Literature Review

Existing Platform	Weakness	Proposed System Benefit
Tazah Technologies Pakistan	Covers only general fresh produce no peanut category, no AI detection, no consumer direct parcel ordering	Peanut specific platform with AI quality detection, disease advisory, and direct consumer parcel delivery
Plantix India/Germany (PEAT GmbH, 135M downloads globally)	Downloaded 135 million times, serving 10 million farmers annually, Focused on Indian crops and conditions.	No seed quality detection, no mandi price tracking, no farmer to consumer marketplace, no peanut-specific model, not available in Urdu voice.
Traditional Mandi System	Farmers sell entirely through arthis with no price visibility and no access to buyers outside local area	direct farmer to consumer connection eliminating middlemen

Seed-X Israel AI-powered computer vision analyzing individual seed size, color, and texture to predict germination rates.	Industrial-scale hardware costing thousands of dollars, accessible only to large seed companies. No smartphone deployment, no peanut specific Pakistani model, no farmer facing interface, no Urdu support, no marketplace integration	First smartphone based peanut seed quality analysis system trained on Pakistani data from NARC and BARI accessible to any farmer in Pothwar at zero hardware cost.
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International Systems

Several international platforms have been developed to assist farmers with crop management and agricultural decision-making. These include:

KissanAI AI-based farming assistant
Plantix Mobile-based plant disease detection platform
CropIn Data-driven crop lifecycle management
AGRIVI Farm management and analytics system
FarmQA Digital agronomic advisory platform
AgriPilot AI Autonomous AI-powered crop monitoring

Pakistan Based Systems

In Pakistan, several digital agriculture initiatives have been launched:

Farmdar Satellite based crop monitoring for Pakistani farmers
Telenor Kissan Dost Mobile based agricultural advisory service
BaKhabar Kissan Digital farming advisory and market price platform
Kisan 360 Integrated farming information and advisory service

3. Problem Statement

Pakistan's peanut sector faces several critical challenges that directly impact productivity, crop health, farmer decision-making, and access to agricultural knowledge.

1. Lack of Scientific Seed Quality Assessment

Farmers plant peanut seeds without any scientific quality evaluation, leading to germination failures ranging from 30 to 40 percent. No affordable and accessible seed testing solution exists at the farm level, forcing farmers to rely on visual inspection and local dealer recommendations. Furthermore, access to certified seed quality standards and improved varieties remains limited.

2. Lack of Early Disease Detection, Disease Progression Prediction, and Preventive Intelligence

Peanut crops in the Pothwar region are vulnerable to diseases such as Early Leaf Spot, Late Leaf Spot, Collar Rot, and Crown Rot. Most farmers identify diseases only after visible symptoms become severe, resulting in delayed intervention and significant crop losses. There is currently no intelligent system capable of providing early disease detection, severity assessment, and preventive guidance at the farm level.

3. Lack of Continuous Intelligent Agricultural Assistance Across the Crop Lifecycle

Farmers face challenges throughout the complete crop lifecycle, including seed selection, crop management, disease control, and cultivation decisions. Existing agricultural solutions are fragmented and focus on individual tasks rather than providing continuous support. Farmers often rely on personal experience, local advice, or delayed expert consultation, resulting in inconsistent decision-making and reduced productivity. Farmers have no intelligent system capable of predicting how a disease may progress over time or providing early warnings before severe crop damage occurs.

4. Research Exists but Limited Market Awareness and Agricultural Knowledge Accessibility

NARC Islamabad and BARI Chakwal possess decades of valuable research related to peanut seed varieties, disease management, cultivation practices, and production techniques. However, this knowledge remains largely confined to research institutions and rarely reaches farmers in a practical and accessible form. Additionally, farmers often lack access to reliable market awareness, buyer information, and data-driven agricultural insights needed for informed decision-making.

4. Proposed Solution

The proposed system introduces **PeanutIQ**, an Agentic AI Powered Seed-to-Harvest Smart Peanut Farming Ecosystem capable of providing intelligent, personalized, and continuous support throughout the peanut crop lifecycle.

1. AI-Based Seed Intelligence System

PeanutIQ provides an AI-powered Seed Intelligence module that analyzes peanut seed images using computer vision and deep learning techniques. The system evaluates seed quality, detects defects, estimates germination potential, generates health scores, and assists farmers in making informed planting decisions before sowing.

2. AI-Based Disease Intelligence System

The Disease Intelligence module enables farmers to upload crop images and receive AI-based disease identification, severity assessment, **disease progression prediction**, outbreak risk analysis, and personalized preventive recommendations. By combining disease severity, crop stage, weather conditions, and historical disease data, the system predicts disease progression and generates early warning alerts before severe crop damage occurs.

3. Agentic AI Smart Farming Companion

PeanutIQ introduces a multi-agent intelligent farming companion that combines farmer history, crop information, disease analysis, and agricultural knowledge to generate personalized and explainable recommendations. The companion provides continuous assistance throughout the crop lifecycle using memory, reasoning, contextual understanding, and voice-based interaction.

4. Research-to-Farmer Knowledge Delivery Platform

PeanutIQ transforms institutional knowledge from NARC and BARI into an accessible digital advisory system. Through Retrieval-Augmented Generation (RAG), trusted agricultural knowledge, cultivation practices, disease management guidelines, and market-awareness information are delivered directly to farmers in a practical and understandable format.

Future Ecosystem Expansion

Future versions of PeanutIQ may include agricultural marketplaces, buyer connectivity, mandi intelligence, direct consumer access, B2B trading modules, and support for additional crops beyond peanuts.

4.1 Objectives

- 1.** Develop a CNN-based Seed Intelligence model capable of analyzing peanut seed images and assessing seed health, defects, quality, and germination potential using smartphone-captured images.
- 2.** Develop a CNN-based Disease Intelligence model capable of detecting major peanut diseases and estimating disease severity levels from crop images.
- 3.** Design and implement an Agentic AI Smart Farming Companion capable of providing personalized, context-aware, and explainable agricultural assistance throughout the peanut crop lifecycle.
- 4.** Develop a memory-enabled recommendation system that utilizes farmer history, crop information, disease analysis, and contextual data to generate intelligent recommendations.
- 5.** Implement a Retrieval-Augmented Generation (RAG) framework to transform NARC and BARI research knowledge into accessible and practical guidance for farmers.
- 6.** Integrate explainable AI mechanisms to provide transparent reasoning behind recommendations and disease-related decisions.
- 7.** Develop a voice-enabled interaction system supporting Urdu and English communication to improve accessibility for farmers.

8. Create a centralized agricultural knowledge platform that bridges the gap between institutional research and on-ground farming practices.
9. Design a scalable Seed-to-Harvest ecosystem that supports continuous decision-making assistance from seed selection to crop management.
10. Evaluate the performance, accuracy, usability, and effectiveness of the proposed AI models and Agentic AI system using real-world peanut farming scenarios.
11. Develop a predictive disease analysis model capable of forecasting disease progression and outbreak risk using disease severity, crop stage, weather conditions, and historical disease records.

Future Objectives

- Agricultural marketplace integration
- Buyer and processor connectivity
- Mandi price intelligence and forecasting
- B2B trading platform
- Consumer direct ordering system
- Farm labour connectivity
- Multi-crop support and expansion

5. Scope

The scope of this project encompasses the design, development, and evaluation of **PeanutIQ**, an Agentic AI Powered Seed-to-Harvest Smart Peanut Farming Ecosystem consisting of Android mobile and web-based platforms. The system is initially targeted toward peanut farmers in the Pothwar region, including Chakwal, Attock, Talagang, and Rawalpindi, where the majority of Pakistan's peanut production is concentrated.

The primary scope of the project focuses on three core components:

Seed Intelligence

The system will provide AI-powered peanut seed quality assessment using computer vision and deep learning techniques. Farmers will be able to analyze seed health, quality, defects, and germination potential through smartphone-captured images before planting.

Disease Intelligence

The system will provide Early disease detection, severity assessment, disease progression prediction, outbreak risk estimation, and preventive advisory. for major peanut diseases using image-based deep learning models. Farmers will receive disease identification results along with preventive guidance and explainable recommendations.

Agentic AI Smart Farming Companion

The system will implement a multi-agent intelligent farming companion capable of providing personalized assistance throughout the crop lifecycle. The companion will combine disease analysis, seed analysis, farmer history, contextual information, and trusted agricultural knowledge to generate explainable and context-aware recommendations.

The platform will support voice and text-based interaction in Urdu and English to improve accessibility for farmers with varying literacy levels.

AI models will be trained using datasets collected from multiple sources, including field-collected peanut seed and crop images, publicly available agricultural datasets, NARC Islamabad resources, and BARI Chakwal research samples.

The system will also incorporate a research-to-farmer knowledge delivery framework that transforms agricultural knowledge from NARC and BARI into practical recommendations using Retrieval-Augmented Generation (RAG) and knowledge-based reasoning.

Out of Scope

The current implementation will not directly manage financial transactions, payment processing, or autonomous agricultural decision-making. Final farming decisions will remain under farmer control.

Future Scope

Future versions of PeanutIQ may include agricultural marketplaces, mandi price intelligence, buyer and processor connectivity, export support modules, farm labour connectivity, consumer ordering services, and support for additional crops beyond peanuts.

6. Modules

6.1 User Management

- Allow users to sign up and log in using phone number or email with OTP verification.
- Enable users to manage profiles including farm location, crop type, language preference, and contact details.
- Support user roles including farmers, researchers, and administrators.
- Implement secure authentication using JWT for protecting user sessions

6.2 Seed Intelligence Module

- Farmer captures or uploads peanut seed batch images through mobile application.
- OpenCV segments individual seeds from batch images for separate analysis.
- CNN model classifies seeds as healthy, damaged, diseased, underdeveloped, or defective.
- Analyze seed size uniformity and identify abnormal seeds.
- Estimate germination potential using visual quality indicators.
- Generate seed health score and overall quality grade.
- Provide recommended actions before planting.

- Generate downloadable PDF quality reports with analysis results.

6.3 Disease Intelligence Module

- Allow farmers to photograph diseased leaves, stems, pods, or entire plants for AI analysis.
- CNN model identifies major peanut diseases from uploaded crop images.
- Detect disease severity levels and progression stages.
- Generate confidence scores for predictions.
- Provide explainable disease analysis and symptom descriptions.
- Generate preventive and management recommendations.
- Maintain disease history for future analysis and monitoring.
- Predict disease progression based on disease severity, crop stage, weather conditions, and historical disease records.
- Generate outbreak risk levels (Low, Medium, High).
- Provide early warning alerts before severe crop damage occurs.

6.4 Agentic AI Farming Companion Module

This module implements the core intelligent assistant capable of reasoning, memory management, and contextual agricultural decision-making throughout the complete peanut crop lifecycle.

Supervisor Agent

- Coordinates all specialized agents.
- Manages workflow execution.
- Generates final responses for farmers.

Memory Agent

- Stores farmer profile information.
- Maintains crop history and disease history.
- Stores previous recommendations and interactions.
- Enables personalized assistance.

Knowledge Agent (RAG)

- Retrieves trusted agricultural knowledge from NARC and BARI sources.
- Retrieves cultivation practices and disease management guidelines.
- Grounds recommendations using validated agricultural information.

Recommendation Agent

- Combines outputs from Seed Intelligence and Disease Intelligence modules.
- Performs contextual reasoning using farmer history and knowledge sources.
- Generates personalized and explainable recommendations.
- Provides crop lifecycle guidance and predictive disease risk analysis.

Validation Layer

- Implements Confidence Thresholds for prediction validation.

- Applies Guardrails to prevent unsafe recommendations.
- Performs Knowledge Base Validation.
- Resolves conflicting outputs through Agent Arbitration.
- Supports Human-in-the-Loop verification when uncertainty exists.

Core Functions

- Context-aware reasoning.
- Continuous agricultural assistance.
- Personalized farmer interaction.
- Explainable AI recommendations.
- Crop lifecycle support from seed selection to harvest.

6.5 Knowledge Base & Research Delivery Module

- Store NARC and BARI peanut cultivation knowledge.
- Store disease management protocols and best practices.
- Store seed quality standards and cultivation guidelines.
- Provide knowledge retrieval support for Agentic AI reasoning.
- Transform institutional research into farmer-friendly recommendations.
- Store historical disease progression records.
- Provide disease forecasting knowledge for predictive analysis.

6.6 Voice Enabled Interaction

- Allow users to interact with the platform using voice commands in Urdu and English.

- Convert speech to text for voice-based system interaction.
- Convert AI responses into natural voice output.
- Deliver disease analysis, seed analysis, and recommendations through voice.
- Support low-literacy farmers through hands-free interaction.
- Enable complete conversational interaction with the Agentic AI Companion.

6.7 Notification Module

- Notify users about recommendation updates and important alerts.
- Send disease outbreak alerts when risk levels increase.
- Generate preventive advisory notifications.
- Send reminder notifications for recommended actions.
- Deliver personalized alerts based on farmer profile and crop stage.
- Notify farmers about predicted disease progression.
- Send outbreak risk alerts.
- Recommend preventive actions before disease severity increases.

6.8 Dashboard & Reporting Module

- Display seed quality analysis reports.
- Display disease detection history and severity records.
- Display recommendation history.
- Provide farmer activity summaries.
- Generate downloadable PDF reports.

- Present crop lifecycle insights and trends.
- Display disease progression trends.
- Display predicted outbreak risk.
- Display disease severity timeline.

6.9 Admin Panel

- Manage user accounts and platform activities.
- Manage knowledge base content and research updates.
- Monitor AI model performance and usage statistics.
- Verify agricultural content before publication.
- Manage reported issues and platform maintenance.
- Monitor overall system activity and user growth.

Future Ecosystem Expansion

- Inputs Marketplace
- Harvest Marketplace
- B2B Dealer Connect
- Consumer Direct Ordering
- Live Mandi Price Intelligence
- Bulk Buyer Inquiry Board
- Farm Labour Connect

- Export Connectivity Platform

- Real-time disease forecasting using weather APIs and satellite data.

7. System Limitations / Constraints

- AI model accuracy depends on image quality. Blurry images, poor lighting, or incorrect camera angles may affect seed and disease detection results.
- The system analyzes only visible features of seeds and plants. Internal defects and laboratory-based quality parameters are outside the project scope.
- Disease detection performance depends on the availability and quality of peanut-specific training datasets.
- The Agentic AI Companion provides decision support and recommendations but does not replace agricultural experts or farmer judgment.
- Voice recognition accuracy may vary due to regional accents, dialects, and background noise.
- Internet connectivity is required for cloud-based AI processing, knowledge retrieval, and real-time advisory services.

8. Data Gathering Approach

Data for this project will be gathered through a structured field-based approach supported by confirmed institutional collaborations and direct agricultural access across the Pothwar region.

NARC Islamabad: Seed bank samples, germination trial records, and disease specimen images will be obtained through a research collaboration confirmed during our field visit in May 2026. NARC's Natural Resources Division will validate seed quality detection outputs and provide institutional support for the project.

Barani Agriculture Research Institute Chakwal: Certified variety samples, disease specimen photographs, and agronomy records specific to Pothwar rainfed conditions will be collected directly from BARI Chakwal and Groundnut Research Station Attock.

Direct Field Access Pothwar Region: The project team has direct and established access across the Pothwar region through long standing agricultural roots in peanut cultivation. The team actively cultivates peanuts in the region, providing firsthand access

to seed samples, plant disease specimens, and real crop conditions throughout the growing season. This gives the project a primary, authentic, Pakistan specific dataset that is unavailable to any other research team.

Farmer Surveys: Structured surveys will be administered through established farmer networks and direct contacts across Pothwar, targeting over 100 respondents covering seed practices, disease challenges, and pricing pain points.

9. Tools and Technologies

Tools / Technologies	Rationale
Flutter	Mobile App Frontend (Android)
React.js	Web Dashboard Frontend
Python (FastAPI)	Backend REST API
PostgreSQL	Database
AWS S3 + EC2	Cloud Storage and Deployment
Docker	Containerization
TensorFlow / Keras	AI Model Training
MobileNetV2	Seed Quality Detection CNN
EfficientNet-Bo	Plant Disease Detection CNN
YOLOv8	Impurity Detection
OpenCV	Image Processing
Scikit-learn	Germination Rate Prediction
JWT + bcrypt	Authentication and Security
TCS / Leopards / Trax APIs	Parcel Booking and Tracking
Speech to Text API	Urdu Voice Interface
XG Boostt /Random forest	